BHUPENDRA SINGH

Associate AI Engineer • GenAI & Agentic Systems • RAG Pipelines • Document Intelligence
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GitHub • Portfolio

B.Tech CSE graduate (PGDM-Data Science & Applied AI) with hands-on experience building end-to-end GenAI systems: Agentic RAG pipelines, LLM-powered document intelligence workflows. Skilled in LangChain, LlamaIndex, OpenAI API, HuggingFace, and FAISS/Pinecone vector stores. Proven track record delivering production-grade Python solutions from automated ETL pipelines to multi-step AI agents with tool-calling, memory management, and retrieval-augmented reasoning. Eager to apply these capabilities to large-scale legal document datasets.

TECHNICAL SKILLS

AI Agents & Orchestration: LangChain Agents, LlamaIndex, Tool-calling, Memory management, ReAct / Plan-and-Execute loops
GenAI & LLM Systems: OpenAI API, HuggingFace Transformers, Prompt engineering, RAG, Agentic RAG, LLM evaluation
Document Intelligence: Document classification, PII detection (spaCy NER, Microsoft Presidio), structured extraction
Vector Stores: FAISS, Pinecone (familiar), Weaviate (familiar), embedding pipelines, retrieval optimisation
ML & Deep Learning: Scikit-learn, TensorFlow, PyTorch, CNNs, LSTM, cross-validation, hyperparameter tuning
Data Engineering: ETL pipelines (500K+ records), PostgreSQL, MySQL, Snowflake (basics), Apache Spark / PySpark (basics)
MLOps & Deployment: Streamlit, Flask, REST APIs, Docker (fundamentals), AWS EC2/Lambda/S3 (basics), CI/CD
Languages & Tools: Python, SQL, Bash (basics), Git/GitHub, Power BI, Jupyter, VS Code

INTERNSHIP EXPERIENCE

AI Engineer Intern — Imarticus Learning (Applied AI Program) Bangalore • 2025 – Present

- Built multi-step LLM agent using LangChain tool-calling to automate LinkedIn data extraction and candidate profiling for 1,000+ professionals, reducing manual effort by 80%.
- Designed Agentic RAG pipeline combining document embedding, FAISS vector indexing, and an AI agent layer to enable intelligent Q&A over structured candidate datasets.
- Implemented PII detection module using spaCy NER to identify and redact sensitive personal data (names, emails, phone numbers) before downstream model ingestion.
- Built modular Python ETL pipelines for scalable data extraction, transformation, and validation supporting AI-driven decision systems.

AI/ML Intern — Vertex Plus Technologies Ltd. Jaipur • May 2025 – Jul 2025

- Built CNN-based computer vision model (InceptionV3 backbone) achieving 95%+ accuracy on pole detection across 10,000+ images; automated preprocessing pipeline cut manual effort by 70%.
- Improved detection efficiency by 60% via hyperparameter optimisation; evaluated model using precision, recall, and F1-score to ensure robust generalisation.

Data Science Intern — Celebal Technologies Jaipur • May 2024 – Aug 2024

- Designed and implemented scalable ETL pipelines handling 500K+ structured records; performed feature engineering and data validation to support downstream ML workflows.
- Built Power BI dashboards that reduced business decision time by 30%; optimised SQL-based extraction improving data processing efficiency and data quality metrics by 20%.

FREELANCE – AI & WEB APPLICATION DEVELOPER

- Architected and deployed AI-powered web applications integrating multi-step LLM agents for automation, document processing, and intelligent workflows.
- Built REST-based backend services for scalable AI model inference, prompt management, and third-party API integrations.
- Developed end-to-end systems (frontend, backend, database, AI orchestration) with retrieval pipelines enhancing contextual response accuracy.

KEY PROJECTS

AI-Based Job Prep & Search Platform : LangChain Agents · FAISS · OpenAI · RAG

- Built GenAI career assistant featuring a multi-step LLM agent with tool-calling: live job search (RapidAPI), document Q&A, and session memory — directly mapping to agentic document intelligence architectures.
- Implemented Agentic RAG pipeline: document embedding → FAISS vector indexing → semantic retrieval → prompt engineering → response evaluation; improved user engagement by 45% in prototype testing.
- Designed modular backend supporting concurrent API requests; added PII redaction layer to sanitise user-uploaded resumes before LLM processing.

Auto EDA Generator : OpenAI API · Pandas · NumPy · Streamlit

- Built automated EDA system generating statistical reports (feature distribution, correlation, anomaly detection) within 15 seconds per dataset; integrated OpenAI API for natural-language insight narration, reducing manual analysis effort by 80%.

Violence Detection : CNN + LSTM Hybrid : InceptionV3 · LSTM · TensorFlow

- Designed CNN+LSTM hybrid achieving 92% accuracy (precision 0.88, recall 0.90); integrated temporal motion-sequence modelling for improved video understanding.

EDUCATION

PGDM Data Science & Applied AI : Imarticus Learning Bangalore · 2025 – Present

B.Tech Computer Science & Engineering : Poornima University Jaipur · 2021 – 2025

CGPA: 6.5 / 10

CERTIFICATIONS

Generative AI with Agentic AI · Udemy

Centre of Excellence in Data Science · Celebal Technologies

Data Science Internship Certificate · Celebal Technologies